

Workshop 3: Plotting

TECH2: Introduction to Programming, Data, and Information Technology

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See GitHub repository for notebooks and data:

<https://github.com/richardfoltyn/TECH2-H26>

Exercise 1: Plotting functions

Consider the following function in one variable x ,

$$f(x) = \sin(20x) + 5x^2$$

which is the sum of the sine function and a quadratic function.

Perform the following tasks:

1. Define a Python function that returns the function value $f(x)$ for some argument x . The function should work with both scalar numbers and NumPy arrays.

Hint: The sine function is implemented in `np.sin()`.

2. Evaluate the function on a uniformly spaced grid of x -values on the interval $[-1, 1]$ using 51 grid points.

Create a plot depicting the function on this interval. Set the x -ticks to the values $[-1, 0, 1]$ and add a legend.

3. Locate the point x_{min} where the function attains its minimum on the grid you created and print the values $(x_{min}, f(x_{min}))$.

Hint: You can use the NumPy function `np.argmin()` for this.

4. Augment the plot you created earlier with a horizontal and vertical line indicating where the minimum is located. Add a black dot at the point $(x_{min}, f(x_{min}))$.

Hint: Horizontal and vertical lines can be added with `axhline()` and `axvline()`. A single dot can be added in various ways, e.g., using `scatter()`.

5. Repeat the exercise but now use $N = 501$ grid points to evaluate the function.

Exercise 2: Visualizing US business cycles and recessions

In this exercise, you are asked to plot two main business cycle indicators, GDP growth and the unemployment rate, for the United States.

1. Load the data from the CSV file `FRED/FRED_quarterly.csv` located in the repository's data folder.

Familiarize yourself with the data by printing the columns present in the data set and the first 5 rows of data.

Hint: Use `read_csv()` with `parse_dates=['DATE']` and `index_col='DATE'` to automatically parse strings stored in the DATE column as dates and set DATE as the index.

2. Compute the quarterly GDP growth (i.e., the relative change of the column GDP vs. the previous quarter) in percent.

Hint: This can be achieved using the `pct_change()` method.

3. Create a figure with two subplots (arranged in 2 rows) which show the GDP growth and the unemployment rate.

Hint: You can construct the graph either using Matplotlib's plotting function or using pandas's `DataFrame.plot()` method.

4. You want to understand how the business cycle indicators behave around recession starts in the US. The National Bureau of Economic Research (NBER) is a research organization that dates US business cycles.

The peak column records the month immediately before each recession starts.

1. Load the business cycle dates from `NBER_cycle_dates.csv` located in the data/ folder.

Drop all rows where the peak date is before 1948.

Hint: Use `pd.read_csv(..., parse_dates=['peak', 'trough'])` to automatically parse strings in the peak and trough columns.

2. Loop through all remaining business cycle peak dates and add a vertical line to each of the two subplots for GDP growth and unemployment at each date. These lines mark the months immediately before recession starts.
3. Can you detect any patterns for GDP growth and the unemployment rate around recession starts?

Exercise 3: Daily returns of US stock market indices

In this exercise, we examine how the three major US stock market indices performed in 2025. Use the data in `indices.csv` from the folder `../data/stockmarket` to solve the following tasks:

1. Load the CSV data and restrict the sample to the calendar year 2025.

Hint: The data is tab-separated, so you need to specify `sep='\t'` when reading in the data.

Hint: You should use `read_csv()` with `parse_dates=['Date']` to automatically parse strings stored in the Date column as dates.

2. The data comes in a "long" format where each date-ticker combination is stored in a separate row. For the following analysis, the data needs to be reshaped such that each ticker symbol is in its own column ("wide" format).

Use the `pivot()` method to reshape the DataFrame into the desired form. Consult the [user guide](#) to see a graphical illustration.

Your pivoted DataFrame should look as follows:

Ticker	DJIA	NASDAQ	SP500
Date			
2025-01-02	42392.3	19280.8	5868.5
2025-01-03	42732.1	19621.7	5942.5

Ticker	DJIA	NASDAQ	SP500
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3. Create a line plot which shows the time series for each of the three ticker symbols in a single graph, for example using `DataFrame.plot()`.
4. The graph you just created is not very satisfactory as the three indices are recorded on vastly different scales. Express each index relative to its initial value in 2025 (so that all three start with the value 1.0) and recreate the previous graph with this normalized data.
5. Another way to check whether the three indices are co-moving is to compute and plot their daily returns. For each index, compute the daily returns, i.e., the relative change vs. the previous closing price in percent (e.g., using `pct_change()`), and plot the three time series of daily returns in a single graph.
6. The previous plots suggest that the three indices co-move a lot.
 1. In order to quantify the strength of this co-movement, compute the pairwise correlations of daily returns using `corr()`.
 2. Create a figure with 3-by-3 subplots where each off-diagonal plot shows the scatter plot for two of the indices.
Hint: You can use the function `scatter_matrix()` to accomplish this task. Alternatively, you can create a figure with 3-by-3 subplots using Matplotlib's `subplots()`, iterate over all rows and columns and add a `scatter()` plot to each axes object.
 3. Add the correlation coefficient for each stock market index pair as text to the corresponding subplot in the figure you created (this can be done using `text()`).

Exercise 4: Population and electric cars in Norway

How do Norwegian municipalities differ in population size, population density, and the use of electric cars? Use the data in `../data/ssb/municipalities.csv` to explore these questions.

The file contains one row per municipality. The variables needed here are:

- `municipality`: Municipality name.
- `population`: Number of residents.
- `land_area`: Land area in square kilometers, excluding freshwater.
- `cars`: Number of registered passenger cars, including electric cars.
- `electric_cars`: Number of electric passenger cars, excluding hybrids.

The data come from Statistics Norway (SSB); see the [data description](#) for details.

1. Load the variables listed above into a `DataFrame`. Display the first five rows and count the municipalities. Are any of the population, land area, or car counts missing?
Hint: Use the `usecols` argument of `pd.read_csv()` to select columns, and `DataFrame.info()` with `show_counts=True` to inspect the number of nonmissing observations.
2. Create a new column `pop_density` containing population divided by land area. This measures residents per square kilometer.

Create a figure with two histograms arranged in one row: population size and population density. Use 30 bins in each histogram. Express population in thousands and label both axes.
Hint: Use `plt.subplots(1, 2)` and the `hist()` method of each Axes object. The height of each bar should show the number of municipalities in that bin.

3. Compute the percentage of passenger cars that are electric in each municipality as $100 * \text{electric_cars} / \text{cars}$, and store it in a new column `ev_share`.

Create a second figure with two histograms arranged in one row: the number of electric cars (in thousands) and the EV share (in percent). Again, use 30 bins and label both axes.

How do these distributions differ? Why might the EV share be more useful than the number of electric cars when comparing municipalities?

4. Create a scatter plot with population density on the horizontal axis and the EV share (in percent) on the vertical axis. Use one equally sized point per municipality and some transparency to make overlapping points easier to see.

Describe the relationship between population density and the EV share.

5. **[Advanced]** Recreate the scatter plot with a logarithmic horizontal axis. Keep the EV share on its original percentage scale.

Hint: Use `set_xscale()` with `'log'`. Equal distances on this axis represent equal ratios: moving from 1 to 10 residents per square kilometer takes the same space as moving from 10 to 100.

Why is this plot easier to read?